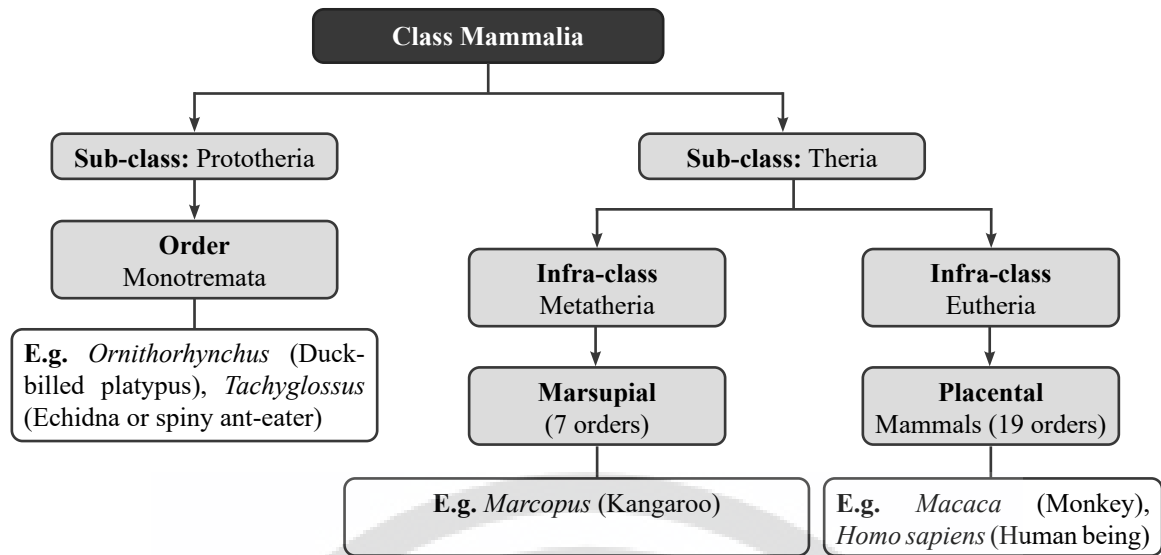


Animal Kingdom

SHORT NOTES

- ❖ **Types of Symmetry:**
 - + **Asymmetrical:** Sponges.
 - + **Radially Symmetrical:** Coelenterates, Ctenophores and Adult Echinoderms.
 - + **Bilaterally Symmetrical:** Annelids, Arthropods, Platyhelminthes, Larva of Echinoderms, Hemichordates and Chordates.
- ❖ **Level of Organisation:**
 - + **Cellular Organisation:** Sponges.
 - + **Tissue Level:** Coelenterates and ctenophores.
 - + **Organ Level:** Platyhelminthes.
 - + **Organ System Level:** Annelids, Arthropods, Molluscs, Echinoderms and Chordates.
- ❖ **Complete Digestive System:** Starts from Aschelminthes to Chordates.
- ❖ **Closed Type of Circulatory System:** Annelids, Cephalopod, molluscs and vertebrates.
- ❖ **Coelom:**
 - + **Acoelomates:** Porifera, Coelenterata, Ctenophora and Platyhelminthes.
 - + **Pseudocoelomates:** Aschelminthes
 - + **Eucoelomates:** Annelids to Chordates.
- ❖ **Segmentation:** Annelids, Arthropods and Chordates.
- ❖ **Porifera (Sponges):** Water transport or canal system.
- ❖ **Echinodermata:** Water vascular system or Ambulacral system.
- ❖ **Coelenterata:** Metagenesis or Alternation of generation or Polymorphism (e.g., *Obelia*).
- ❖ **Ctenophores:** Bioluminescence and all organisms are marine.
- ❖ **Platyhelminthes:** Hooks and suckers are present in parasitic forms.
- ❖ **Aschelminthes:** Muscular pharynx, male (smaller + curved) and female (longer + straight).
- ❖ Parapodia (locomotory organ) of aquatic annelids like *Nereis* which help in swimming.
- ❖ *Nereis* is dioecious but earthworms and leeches are monoecious.
- ❖ **Arthropoda:** Largest phylum.
- ❖ **Mollusca:** 2nd Largest phylum.
- ❖ **Arthropoda:** Chitinous exoskeleton & Blood is colourless (Haemolymph).
- ❖ **Tracheal system:** Respiratory organ in insects.
- ❖ **Mollusca:** Presence of (ctenidia + organ of Bojanus + pearl formation).
- ❖ **Neopilina:** Connective link between Annelida and Mollusca.
- ❖ Hemichordata is the connecting link between nonchordata & chordata.
- ❖ **Central Nervous System:**
 - + **Non-chordates:** (Solid + Double + Ventral side of body)
 - + **Chordates:** (Hollow + Single + Dorsal side of body)
- ❖ **Urochordata:** Retrogressive metamorphosis.
- ❖ **Cephalochordates:** Progressive metamorphosis.
- ❖ **Heart Chambers:**
 - + Pisces-2; Amphibians-3; Reptiles-3 (**crocodiles-4 exception**); Aves-4; Mammals-4.
- ❖ **Chimaera:** Connecting link between chondrichthyes and osteichthyes.
- ❖ **Chondrichthyes:** Cartilaginous Fishes & marine.
- ❖ **Osteichthyes:** Bony fishes & Both marine and fresh water fishes.
- ❖ **Reptiles:**
 - + Exoskeleton is made up of horny epidermal scales or scutes.
 - + Skin is dry, cornified, rough and non-glandular.
- ❖ **Aves:**
 - + Feathers present.
 - + Forelimbs modified into wings.
 - + Hind limbs have scales and are modified for walking, swimming or clasping.
 - + Pneumatic bones.
- ❖ **Mammalia:**
 - + Presence of hair & mammary glands.
 - + External ears or pinnae are present.
 - + Homoiothermic - Aves, mammals

Classification of Mammals



Phylum	Level of Organisation	Symmetry	Coelom	Segmentation	Digestive System	Circulatory System	Respiratory System	Distinctive Features
Porifera	Cellular	Various	Absent	Absent	Absent	Absent	Absent	Body with pores and canals in walls.
Coelenterata (Cnidaria)	Tissue	Radial	Absent	Absent	Incomplete	Absent	Absent	Cnidoblasts present.
Ctenophora	Tissue	Radial	Absent	Absent	Incomplete	Absent	Absent	Comb plates for locomotion.
Platyhelminthes	Organ & Organ-system	Bilateral	Absent	Absent	Incomplete	Absent	Absent	Flat body, suckers.
Aschelminthes	Organ-system	Bilateral	Pseudo-coelomate	Absent	Complete	Absent	Absent	Often worm-shaped, elongated.
Annelida	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Absent	Body segmentation like rings.
Arthropoda	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Present	Exoskeleton of cuticle, jointed appendages.
Mollusca	Organ-system	Bilateral	Coelomate	Absent	Complete	Present	Present	External skeleton of shell usually present.
Echinodermata	Organ-system	Radial	Coelomate	Absent	Complete	Present	Present	Water vascular system, radial symmetry.
Hemichordata	Organ-system	Bilateral	Coelomate	Absent	Complete	Present	Present	Worm-like with proboscis, collar and trunk.
Chordata	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Present	Notochord, dorsal hollow nerve cord, gill slits with limbs or fins.